Database Assignment 4 solutions

1. select e.id, e.name, e.dept_name, d.id, d.name, d.no_of_employees
   -> from Employee as e
   -> cross join Department as d;

2. SELECT e.name, d.name
   -> FROM Employee AS e
   -> INNER JOIN Department AS d
   -> ON e.dept_name = d.name;

3. select *
   -> FROM Employee AS e
   -> left outer join
   -> department as d
   -> on e.dept_name = d.name;

4. select *
   -> FROM Employee AS e
   -> right outer join
   -> department as d
   -> on e.dept_name = d.name;

5. select * from employee
   -> natural join
   -> department;

6. select * from project
   -> cross join
   -> employee;

7. create table EmpBackup like employee;

8. INSERT INTO empbackup
   -> SELECT * FROM employee;

9. create table ProjectArchive like project;

```
insert into ProjectArchive
  -> select * from Project;
```

10. create table Emp (EmpID int auto_increment primary key, EmpName VARCHAR(50), Salary Decimal (10,2));

11. select * from emp where salary < (select avg(salary) from emp);

12. select * from employee where dept_name in (select dept_name from dep where location = 'Mumbai');

13. select * from employee where dept_name in (select dept_name from department where no_of_employees > 5);

14. select max(salary) as max_salary from department where dept_name in (select dept_name from employee);

15.
```
SELECT d1.dept_name AS department1,
    ->      d2.dept_name AS department2,
    ->      d1.location
    -> FROM department d1
    -> JOIN department d2
    -> ON d1.location = d2.location
    -> AND d1.dept_name < d2.dept_name;
```